

Multilevel Models for Airbnb Ratings

An implementation and extension of Chapter 13 of *Statistical Rethinking*

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1 The Problem: A Rating Is Only as Precise as Its Review Count

A listing with one review rated 5.0 is not better than a listing with four hundred reviews averaging 4.83. The raw average treats them as equals, and that is wrong, because the two numbers are not measured with the same precision. This rating issue is common in many customer review based systems and is often taken as a pedagogical example of exploiting Bayesian inference tools to improve estimates and even make purchasing decisions (see 3Blue1Brown's YouTube video [1]).

The data used in this report is taken from the *Inside Airbnb* dataset[2], showing some interesting statistics of London listings. The dataset used is `listings.csv` from the London June 2026 listings (click here to download easily). Across the 70,708 usable London listings, the *mean* rating barely moves with the number of reviews, while the *spread* collapses:

reviews	listings	mean rating	SD of rating
1-2	13,995	4.57	0.865
3-5	10,615	4.66	0.496
6-10	9,959	4.70	0.341
11-50	24,332	4.74	0.239
50+	11,807	4.77	0.179

The spread shrinks by a factor of almost five while the mean moves by two tenths of a point. What we are seeing is sampling noise: an average over two reviews scatters because it is an average over two reviews - as simple as that.

This is exactly the situation Chapter 13 is about. The chapter's answer is that each listing should be estimated using *all* the data, not only its own, and that the amount of information a listing borrows from the others should depend on how little evidence it has of its own. In this report, we will showcase this principle using the above Airbnb data.

2 Reviews as Trials: The Reed-Frog Model in Disguise

Suppose each review comes from a customer who either had a positive experience or did not. Let p_i be the probability that an arbitrary customer has a positive experience at listing i , and let the overall rating be the fraction of positive experiences, rescaled from $[0, 1]$ to the star range $[1, 5]$. Then from the observed rating y_i and the review count N_i we can recover a count of positive reviews:

$$p_i^{\text{obs}} = \frac{y_i - 1}{4}, \quad P_i = \text{round}(p_i^{\text{obs}} N_i).$$

P_i is a count of successes out of N_i trials. That is the tadpole data of Chapter 13, with **listings in place of tanks** and **reviews in place of tadpoles**! Every listing contributes exactly one row, exactly as every tank does.

Rethinking: Average rating vs. success ratio

The important assumption that is being made here is that `review_scores_rating` is an average of one-to-five star ratings, not a proportion of positives. treating it as one is a modelling choice, used to create a complete analogy to the reed frog model.

The analysis below uses 200 listings, 40 drawn from each of the five review count bands above, carrying 5,563 reviews between them with a median of 8 per listing.

3 Three Models, One Question: How Much Should Listings Share?

The three models below differ in exactly one respect: how much one listing's intercept is allowed to know about the others'. That single axis - from *everything* to *nothing* to *as much as the data support* - is the whole content of Chapter 13.

Complete pooling. One intercept for all two hundred listings at once. The model asserts that listings do not differ, and that every gap between them is sampling noise:

$$P_i \sim \text{Binomial}(N_i, p_i), \quad \text{logit}(p_i) = \alpha, \quad \alpha \sim \text{Normal}(0, 1.5).$$

No pooling. The opposite extreme. Every listing gets its own intercept, and the intercepts know nothing about each other (according to Ch. 13, this model has *Amnesia*):

$$P_i \sim \text{Binomial}(N_i, p_i), \quad \text{logit}(p_i) = \alpha_{\text{listing}[i]}, \quad \alpha_j \sim \text{Normal}(0, 1.5).$$

Partial pooling. In between: the prior over the intercepts is itself estimated from the data:

$$P_i \sim \text{Binomial}(N_i, p_i), \quad \text{logit}(p_i) = \alpha_{\text{listing}[i]}, \\ \alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma), \quad \bar{\alpha} \sim \text{Normal}(0, 1.5), \quad \sigma \sim \text{Exponential}(1).$$

Rethinking: Regarding effective number of parameters

Although the second model has two more parameters than the first, it will soon turn out to be the less flexible of the two, and that apparent contradiction is the whole lesson of Ch. 13 - instructive models can be more flexible than their more naive cousins, despite actually having more parameters.

Model names below follow the book. In the code they are spelled out: `m13.0` is `complete_pooling`, `m13.1` is `no_pooling`, `m13.1b` is `no_pooling_flat`, and `m13.2` is `partial`, all defined in `src/03-models.R` in the attached code.

```
m13.0 <- ulam(  
  alist(  
    P ~ dbinom( N , p ),  
    logit(p) <- a,  
    a ~ dnorm( 0 , 1.5 )  
  ), data=dat , chains=4 , log_lik=TRUE )  
  
m13.1 <- ulam(  
  alist(  
    P ~ dbinom( N , p ),  
    logit(p) <- a[listing],  
    a[listing] ~ dnorm( 0 , 1.5 )  
  ), data=dat , chains=4 , log_lik=TRUE )
```

```
m13.2 <- ulam(
  alist(
    P ~ dbinom( N , p ),
    logit(p) <- a[listing],
    a[listing] ~ dnorm( a_bar , sigma ),
    a_bar ~ dnorm( 0 , 1.5 ),
    sigma ~ dexp( 1 )
  ), data=dat , chains=4 , log_lik=TRUE )
```

A sensitivity check on the fixed prior. The Normal(0, 1.5) that m13.0 and m13.1 use is the book's, but it is not weak on this scale. Ratings here sit near 4.79, which means $p \approx 0.947$ and $\text{logit}(p) \approx 2.9$: the prior places the typical listing almost two standard deviations from its own mean, and so pulls every listing toward $p = 0.5$. To check that the comparison below is not an artifact of that, a fourth fit m13.1b was run, identical to m13.1 but with $\alpha_j \sim \text{Normal}(0, 5)$, which really is flat.

4 Fitting the Model: The Funnel and the Non-Centered Fix

Our case's Devil's Funnel. The multilevel model is harder to sample than it looks. The difficulty is that σ and the intercepts are entangled: when σ is small, the α_j must be packed tightly around $\bar{\alpha}$, and when σ is large they may spread out. The posterior therefore has the shape of a funnel, narrow at one end and wide at the other, and a step size that suits one end of the funnel is wrong for the other (in terms of maximizing the runtime efficiency). **This is the precise problem discussed in Figure 13.5 in the book** and illustrated amazingly in Lecture 14 of McElreath's Statistical Rethinking playlist (video link [5]).

The standard remedy, and the one the chapter gives, is to *reparameterize*. Instead of sampling the intercepts directly, sample standardized offsets $z_j \sim \text{Normal}(0, 1)$ and rebuild the intercepts from them. This is the **Non-centered parameterization** discussed in chapter 13.4:

$$\alpha_j = \bar{\alpha} + z_j \sigma.$$

This is the same model - it has the same posterior over α_j . However, the parameters the sampler actually moves through, z_j , no longer change shape when σ changes, and the funnel goes away! (As for the divergent transition effects, we were lucky to not see any of those here, see note below the code box).

```
m13.2nc <- ulam(
  alist(
    P ~ dbinom( N , p ),
    logit(p) <- a_bar + z[listing]*sigma,
    z[listing] ~ dnorm( 0 , 1 ),
    a_bar ~ dnorm( 0 , 1.5 ),
    sigma ~ dexp( 1 ),
    gq> vector[listing]:a <- a_bar + z*sigma
  ), data=dat , chains=4 , log_lik=TRUE )
```

Both versions were fitted. The results from applying `precis()` function from the `rethinking` package:

	divergent	min E-BFMI	max $\hat{R}$	min n_{eff}
centered	0	0.216	1.023	150
non-centered	0	0.736	1.001	714

Neither version produced a single divergent transition, so the warning most often associated with the funnel never fired. This is worth being honest about: the textbook symptom did not appear on this data set (maybe the step size used was small enough). The funnel showed up anyway, in two quieter places. Stan warned that the energy diagnostic E-BFMI was below 0.3 on all four chains of the centered model — it came in at 0.216, against 0.736 after reparameterizing - and the centered model extracted only 150 effectively independent draws of σ from the same 2,000 samples that gave the non-centered model 714. Reparameterizing bought nearly a $\times 5$ improvement in the precision of σ for no extra computation, which will soon be explained through the trace & "trank" (trace-rank as McElreath calls them, liked that nickname) plots.

4.1 Reading the sampler diagnostics

What the diagnostics mean

Divergent transitions. Stan simulates a physical trajectory across the posterior (read my appendix!). When the surface curves more sharply than the step size can follow, the simulated trajectory diverges from the true one, and Stan records it. A divergence means the sampler could not go where it was trying to go, so that region of the posterior is under-explored, and most of the times that means that the fitting result of the algorithm should be discarded.

E-BFMI (Energy Bayesian Fraction of Missing Information):

$$\text{E-BFMI} = \frac{\sum_{m=1}^M (E_m - E_{m-1})^2}{\sum_{m=1}^M (E_m - \bar{E})^2}$$

Measures the ratio of step-to-step energy change variance to total marginal energy variance along the trajectory ($E_m = H(q_m, p_m)$). It quantifies how rapidly the sampler explores different Hamiltonian energy levels. values below 0.3 signal sluggish energy diffusion trapped by narrow geometry (funnels).

$\hat{R}$ (Potential Scale Reduction Factor):

$$\hat{R} = \sqrt{\frac{W + \frac{1}{N}(B - W)}{W}}, \quad W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2$$

Compares between-chain variance (B) to within-chain variance (W) across M chains of length N . Values $\hat{R} \rightarrow 1.0$ (threshold < 1.01) confirm that independent chains initialized from different points have converged to the same distribution.

n_{eff} / ESS_{bulk} (Effective Sample Size):

$$n_{\text{eff}} = \frac{S}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

Adjusts total MCMC samples S for autocorrelation ρ_k at lag k between successive iterations. It calculates the equivalent number of completely independent draws, directly measuring the actual statistical precision delivered by the Markov chain. The last two quantities are explained intuitively in [3] around minute 58:00.

4.2 Trace plots: what a healthy chain looks like

A trace plot draws each of the Markov chain's path over the iterations (e.g. the graph that records the value of each parameter across the iterations). What we want aim to see is what McElreath calls "a fuzzy caterpillar": the chains should sit in the same horizontal band, wiggle rapidly, and none of them should wander off or stick in one place for a stretch. If that happens, that means that the sampling algorithm successfully explores the posterior distribution, intuitively explained by arguments of ergodicity (more details in the appendix :).

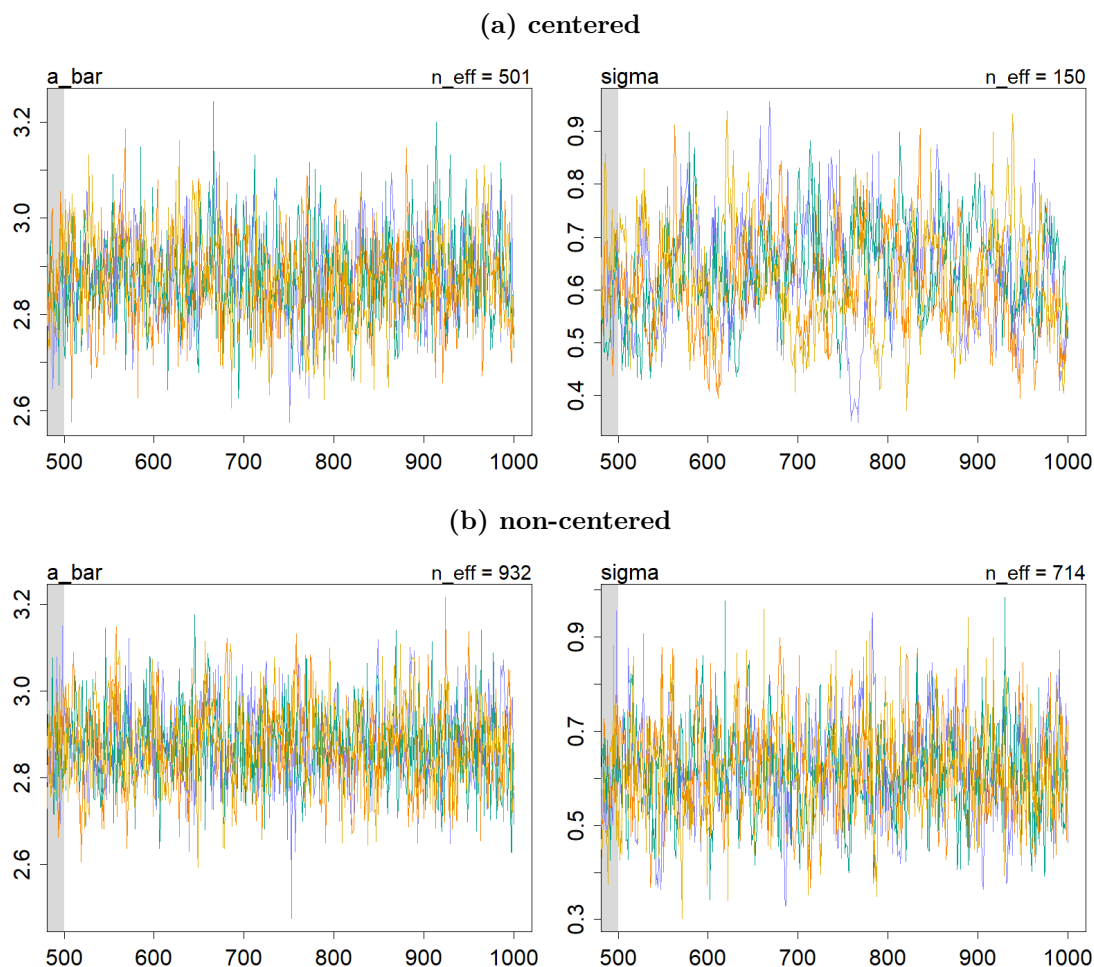


Figure 1: Trace plots for $\bar{\alpha}$ and σ , warmup excluded. Both are stationary (neither is broken), but they are not equally good. The centered σ trace makes slow excursions that last tens of iterations, wandering low for a stretch and then high. The non-centered trace oscillates rapidly with no such drift. Slow excursions mean successive draws carry nearly the same information, which is exactly what a low effective sample size reports.

4.3 Trace-rank plots: the same problem, seen better

Trace plots become hard to read once four chains overlap, because the ink piles up and a chain that is slightly wrong is hidden behind three that are right. A trace rank plot fixes this. All the samples from all the chains are ranked together, from lowest to highest, and then each chain's ranks are histogrammed separately.

The logic is simple. If every chain is exploring the same distribution, then each chain should own roughly its fair share of the low ranks, the middle ranks, and the high ranks. So the histograms should be flat and should overlap. A chain that is stuck somewhere too long will own too many ranks in one region and show up as a histogram that rises above the others.

The warm-up stage in trace plots

Stan discards the first W iterations of each chain to give the sampler time to find the typical set before collecting samples for inference. The default warmup length is $W = 2000$, but this can be adjusted using the `warmup` argument in Stan (you can see in `src/00-setup.R` and `src/04-fit.R` that we use the default value for `warmup` and set `CHAINS=4`).

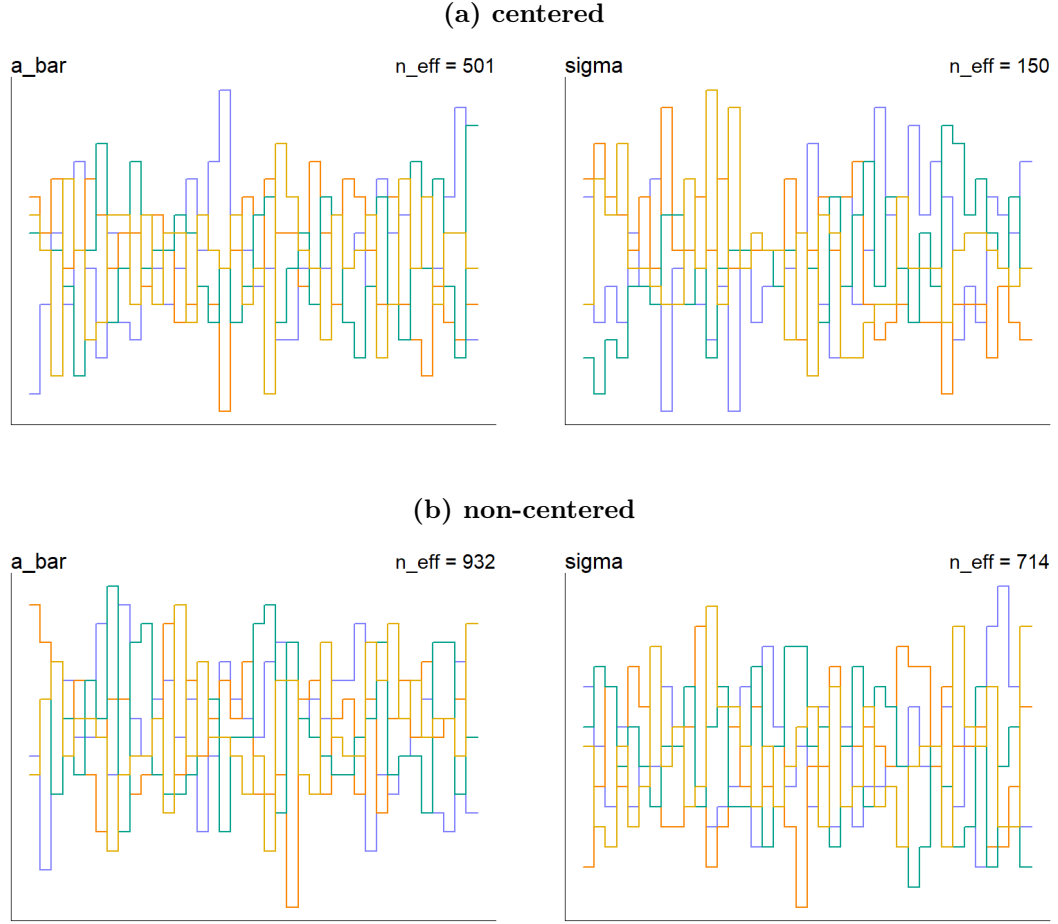


Figure 2: Trace rank plots. Look at the σ panel of (a): one chain holds far more than its share of the lowest ranks at the left edge, and another spikes at the highest ranks on the right. Those are chains that spent too long in one region - the same slow excursions the trace plot showed, now visible as an imbalance rather than as a wiggle. In (b) the four histograms interleave much more evenly.

5 What Partial Pooling Buys: A Population, Not Just 200 Intercepts

A no-pooling model hands back two hundred numbers. A partial-pooling model hands back two hundred numbers *and the distribution they were drawn from*, and that distribution is what lets it say anything at all about a listing it has never seen. This section reads it off in three steps: the two parameters that describe the population, the population they imply, and what that population does to the individual estimates.

5.1 The estimated population: $\bar{\alpha}$ and σ

Reading Table 1: **mean** and **sd** describe the marginal posterior of each parameter, and the two percentage columns give the 89% compatibility interval. The last two columns are the diagnostics from the box above, and both are healthy here.

The numbers themselves say two things. $\bar{\alpha} = 2.88$ on the log-odds scale is $\text{logit}^{-1}(2.88) = 0.947$ as a probability, or $1 + 4(0.947) = 4.79$ on the star scale. That is the average listing, and it matches the population mean of the full data set. And $\sigma = 0.62$ says how far listings genuinely

	mean	sd	5.5%	94.5%	$\hat{R}$	n_{eff}
a_bar	2.876	0.094	2.730	3.031	1.001	932
sigma	0.617	0.103	0.456	0.786	1.001	714

Table 1: `precis(m13.2, depth=1)`. The 200 listing intercepts are hidden (`depth=2` would show them all, thanks to Claude familiarity with coding in R)

differ from one another once sampling noise has been accounted for. It is not zero, so listings really do vary; it is also well below 1.5, which is the fixed prior the no-pooling model was using — the model has decided that listings are four times more alike than that prior assumed, and everything below follows from that one number.

5.2 From log-odds to stars: the population on both scales

Those two numbers are not just summaries; together they *are* the estimated population, $\text{Normal}(\bar{\alpha}, \sigma)$, and we can draw it.

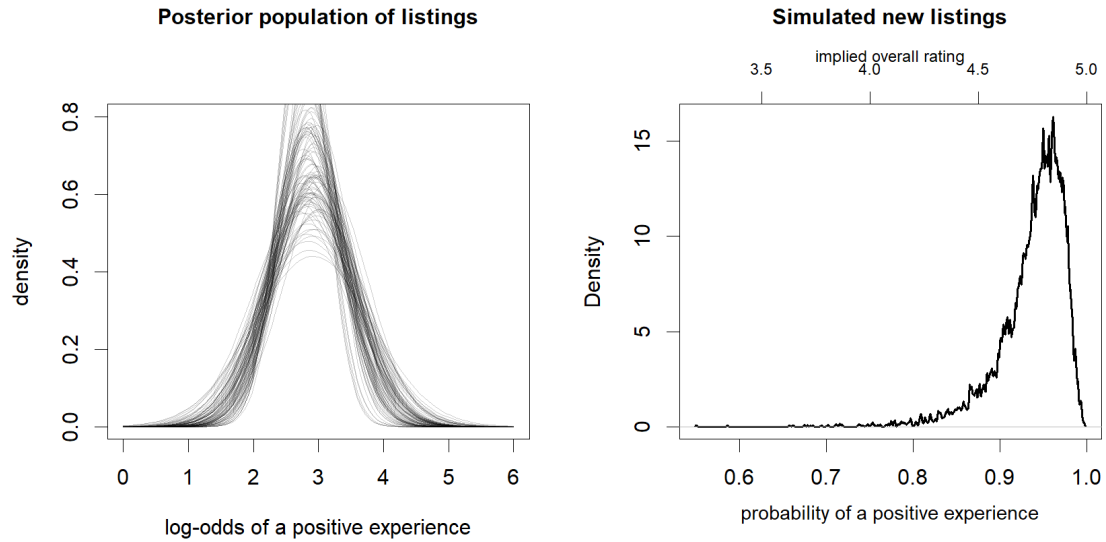


Figure 3: Left: one hundred draws of the population distribution of listings on the log-odds scale, each curve a $\text{Normal}(\bar{\alpha}, \sigma)$ using one posterior sample. The spread of the curves is uncertainty about the population; the width of each curve is the variation among listings. Right: 8,000 simulated listings pushed through the inverse logit, so the horizontal axis is the probability of a positive experience, with the implied star rating marked on top.

```
post <- extract.samples( m13.2 )
plot( NULL , xlim=c(0,6) , ylim=c(0,0.8) ,
      xlab="log-odds of a positive experience" , ylab="density" )
for ( i in 1:100 )
  curve( dnorm(x,post$a_bar[i],post$sigma[i]) , add=TRUE ,
         col=col.alpha("black",0.2) )

sim_listings <- rnorm( 8000 , post$a_bar , post$sigma )
dens( inv_logit(sim_listings) , lwd=2 , adj=0.1 )
```

The transformation matters. On the log-odds scale the population is a symmetric bell. On the probability scale it is strongly skewed, piled up near 1 with a long tail reaching down. That asymmetry is not something we assumed; it is what a symmetric distribution becomes once it

is squeezed through the inverse logit near the top of its range. It is also a fair description of Airbnb: most listings are fine, and the interesting variation is all on the downside.

5.3 Shrinkage: the pooling goes where the evidence is thin

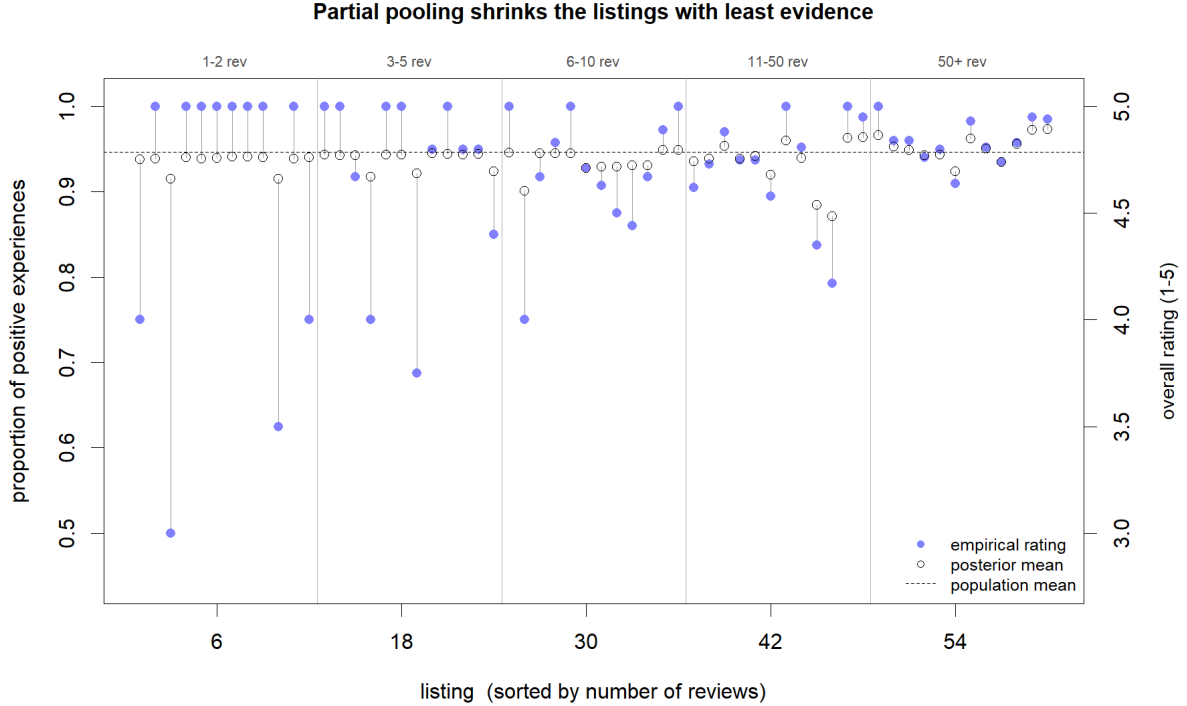


Figure 4: Filled points are the empirical ratings, open points the posterior means, and the segment between them is what pooling did. The dashed line is the population mean $\text{logit}^{-1}(\bar{\alpha})$. Listings are sorted by review count, with the bands marked along the top.

The picture is the one Chapter 13 promises. On the left, where listings have one or two reviews, the segments are long and every one of them points toward the dashed line. On the right, where listings have fifty reviews or more, the open and filled points sit on top of each other. Nothing was done to the model to produce this gradient. It follows from the binomial likelihood: a count out of 400 trials is sharp evidence and overwhelms the prior, while a count out of one trial is almost no evidence and the prior wins.

reviews	mean shift	spread before	spread after	shrink factor
1–2	0.327	0.314	0.032	0.899
3–5	0.396	0.445	0.059	0.869
6–10	0.214	0.257	0.045	0.826
11–50	0.111	0.168	0.063	0.625
50+	0.047	0.116	0.070	0.395

Table 2: All quantities in rating points. “Spread” is the mean distance from the population mean, before and after pooling; the shrink factor is $1 - \text{after/before}$.

One column in that table is not monotone and the reason is instructive. The mean shift for one-to-two-review listings (0.327) is *smaller* than for three-to-five (0.396). This is not an error. A listing with a single review is almost always rated exactly 5.0, and 5.0 is only about 0.2 rating points above the population mean of 4.79, so there is very little distance to travel even though

nearly all of it is taken away. The shrink factor, which measures the *fraction* of the gap that pooling closes, is the honest summary, and it falls monotonically from 0.90 to 0.40.

6 Comparing the Three Models I: Which Predicts Best?

	params	WAIC	SE	dWAIC	dSE	pWAIC
m13.2 partial pooling	202	485.1	28.4	0.0	–	37.5
m13.1b no pooling, flat	200	508.3	25.7	23.2	14.7	73.3
m13.0 complete pooling	1	547.5	42.6	62.4	22.5	2.8
m13.1 no pooling, book	200	690.2	18.3	205.1	18.7	112.1

Table 3: `compare(m13.0, m13.1, m13.1b, m13.2)`. `params` is the actual number of parameters, not the effective one.

WAIC, pWAIC, and dWAIC

WAIC (Watanabe–Akaike Information Criterion):

$$\text{WAIC} = -2 (\text{lppd} - p_{\text{WAIC}}), \quad \text{lppd} = \sum_{i=1}^N \ln \left(\frac{1}{S} \sum_{s=1}^S p(y_i | \boldsymbol{\theta}^{(s)}) \right)$$

Estimates out-of-sample predictive accuracy on the deviance scale (lower is better). It evaluates the log posterior predictive density (lppd) fitted over N observations across S posterior draws and penalizes for flexibility via p_{WAIC} .

p_{WAIC} (Effective Number of Parameters):

$$p_{\text{WAIC}} = \sum_{i=1}^N V_i, \quad V_i = \text{Var}_{s=1..S} \left(\ln p(y_i | \boldsymbol{\theta}^{(s)}) \right)$$

Calculates model complexity by summing the variance of log-likelihood across posterior draws for each observation y_i . Parameters constrained by tight priors or data pooling contribute little, measuring how many parameters the model is effectively *using*, similar to the concept utilized by principal component analysis (PCA) that we’ve studied with Kavitzky.

ΔWAIC (dWAIC) and dSE (Difference and Standard Error):

$$\Delta\text{WAIC}_m = \text{WAIC}_m - \text{WAIC}_{\min}, \quad \text{dSE} = \sqrt{N \cdot \text{Var}_i (\text{WAIC}_{m,i} - \text{WAIC}_{\min,i})}$$

Measures the predictive gap between model m and the best-performing model ($\text{WAIC}_{\min}$). dSE computes the standard error of that point-by-point difference, showing whether a performance gap is statistically significant.

The pWAIC column is the point of the whole exercise. The multilevel model has 202 parameters. The no-pooling models have 200. Yet the multilevel model uses the equivalent of 37.5 of them, against 73.3 and 112.1. It is the model with the most parameters and the least overfitting.

The reason is the adaptive prior. In **m13.1** each intercept is regularized by a fixed $\text{Normal}(0, 1.5)$ that was chosen before seeing any data. In **m13.2** the model learns from the data that the true spread is $\sigma = 0.62$, which is far tighter, and applies that. Regularization the model works out for itself beats regularization we guessed in advance.

The sensitivity fit is worth a sentence, because it comes out backwards from what one might expect. **m13.1b**, with the *wider* prior, has the *lower* pWAIC of the two no-pooling models. The

reason is that $\text{Normal}(0, 1.5)$ is not merely tight, it is tight *in the wrong place*: it is centered on $p = 0.5$ while the data sit at $p = 0.947$. Prior and likelihood pull against each other for every listing, and that conflict inflates the variance of the log-likelihood, which is what pWAIC measures. The comparison survives either way — 37.5 beats both — so the multilevel model’s advantage is not an artifact of the prior we happened to choose.

Complete pooling, `m13.0`, sits at the other extreme. Its pWAIC of 2.8 is the smallest in the table, and it is the *only* model whose effective parameter count is close to its actual one — with one parameter there is nothing for the data to overfit. It still loses to partial pooling by 62 WAIC, because the listings genuinely do differ: $\sigma = 0.62$ is small but it is not zero.

So the table brackets the answer. Complete pooling is too rigid and underfits; no pooling is too free and overfits; partial pooling sits between them and beats both. That ordering is what the whole chapter is about — but WAIC only *estimates* out-of-sample accuracy, and it estimates it from the same data the models were fitted to. The next section repeats the comparison in a setting where nothing has to be estimated, because the truth is known by construction.

7 Comparing the Three Models II: Which Recovers the Truth?

WAIC ranked the models by how well they are expected to predict. That is a useful ranking, but it is an inference: nowhere in the table above does a true value appear. In real data the true probability p_i^{true} behind each listing is unobservable — we see only the noisy outcome $P_i \sim \text{Binomial}(N_i, p_i)$ — so “partial pooling is closer to the truth” is a claim we cannot check directly on Airbnb.

Simulation removes that obstacle. If we *generate* the listings ourselves, we know every p_i^{true} , and we can ask the sharper question directly: which of the three models comes closest to the numbers that actually produced the data? This is the controlled study of Section 13.2 of *Statistical Rethinking* (R code 13.9–13.20), run here with the parameters our own model estimated.

We simulate $K = 60$ listings, divided into four review-count groups of 15 listings each with $N_i \in \{2, 5, 15, 50\}$. We examine two distinct data-generating scenarios:

1. **Fitted Airbnb scenario** ($\bar{\alpha} = 2.88, \sigma = 0.62$): Parameters estimated from the London dataset, characterized by high overall satisfaction ($\bar{p} \approx 0.947$) and tight population concentration ($\sigma = 0.62$).
2. **Textbook scenario** ($\bar{\alpha} = 1.50, \sigma = 1.50$): Parameters from McElreath’s tadpole simulation, characterized by substantial population variation ($\sigma = 1.50$).

For each simulated listing i , true log-odds $\alpha_i^{\text{true}} \sim \text{Normal}(\bar{\alpha}, \sigma)$ are drawn, giving true proportions $p_i^{\text{true}} = \text{logit}^{-1}(\alpha_i^{\text{true}})$, followed by observed counts $P_i \sim \text{Binomial}(N_i, p_i^{\text{true}})$. We fit complete pooling ($\hat{p}_i = \sum P_i / \sum N_i$), no pooling ($\hat{p}_i = P_i / N_i$), and partial pooling (posterior mean $\hat{p}_i = \mathbb{E}[\text{logit}^{-1}(\alpha_i) \mid \text{data}]$), measuring performance by the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{K} \sum_{i=1}^K |\hat{p}_i - p_i^{\text{true}}|.$$

7.1 What the simulation shows

The simulation highlights two critical theoretical insights:

1. **Adaptive trade-off in low vs. high variance regimes**: When population variance is tight ($\sigma = 0.62$, Fitted scenario), complete pooling is surprisingly competitive (MAE = 0.0260) because listings are genuinely similar — it is in fact the better estimator for the

Scenario	Reviews (N_i)	Complete Pooling	No Pooling	Partial Pooling
Fitted Airbnb ($\bar{\alpha} = 2.88, \sigma = 0.62$)	2	0.0296	0.1122	0.0320
	5	0.0310	0.0718	0.0289
	15	0.0166	0.0447	0.0167
	50	0.0269	0.0318	0.0198
	Overall	0.0260	0.0651	0.0243
Textbook ($\bar{\alpha} = 1.50, \sigma = 1.50$)	2	0.1913	0.2097	0.1179
	5	0.2021	0.1234	0.1081
	15	0.1117	0.0725	0.0592
	50	0.1758	0.0526	0.0525
	Overall	0.1702	0.1145	0.0844

Table 4: Mean Absolute Error against ground truth p_i^{true} across 60 simulated listings (15 per review count group); the lowest error in each row is bold. Partial pooling achieves the lowest *overall* error in both scenarios, but not every row: in the Fitted scenario complete pooling edges it out at $N_i = 2$ and ties it at $N_i = 15$, which is the tight- σ effect discussed below.

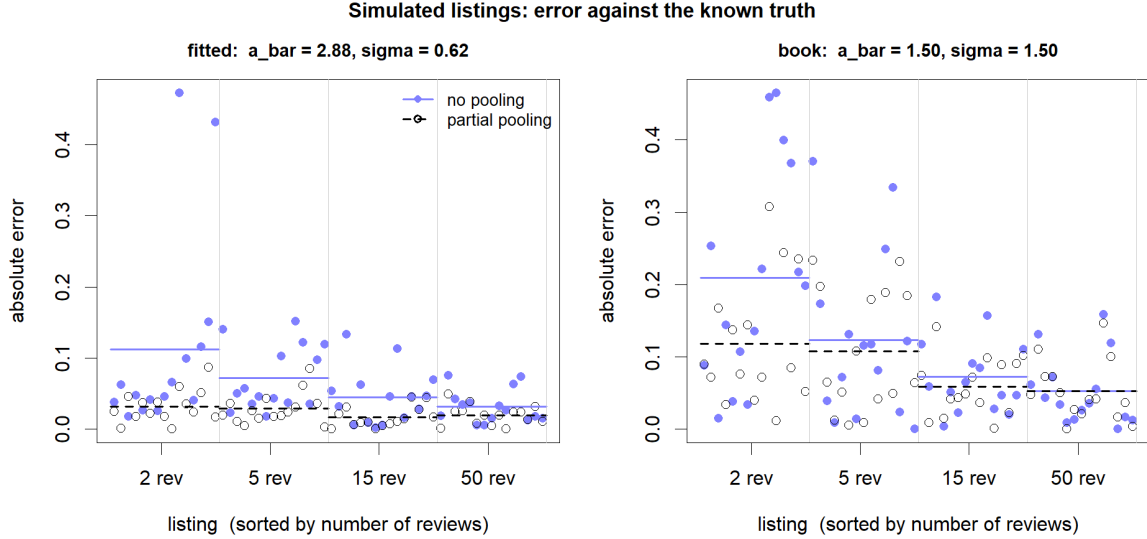


Figure 5: Absolute error for each simulated listing against ground truth p_i^{true} . Blue points and solid lines represent no pooling; open circles and dashed black lines represent partial pooling. Left: Fitted Airbnb scenario ($\bar{\alpha} = 2.88, \sigma = 0.62$). Right: Textbook scenario ($\bar{\alpha} = 1.50, \sigma = 1.50$). Partial pooling prevents large errors in low-review listings while matching no-pooling performance when evidence ($N_i = 50$) is high.

sparsest listings, beating partial pooling at $N_i = 2$ (0.0296 against 0.0320). Partial pooling still wins overall (MAE = 0.0243) by allowing listings with large sample sizes ($N_i = 50$) to deviate appropriately. When population variance is high ($\sigma = 1.50$, Textbook scenario), complete pooling fails severely (MAE = 0.1702) due to underfitting, and partial pooling wins in every group. The lesson is that partial pooling is never much worse than the better of the two extremes, while each extreme is badly wrong in one of the two regimes.

2. **Protection against sampling noise for low N_i :** For listings with only 2 reviews ($N_i = 2$), no pooling suffers catastrophic estimation errors (MAE = 0.1122 in Fitted, 0.2097 in Textbook) because a single binomial outcome produces extreme estimates (0.0, 0.5, or 1.0).

Partial pooling shrinks these estimates toward $\bar{\alpha}$, reducing the error for $N_i = 2$ by nearly 70% in the Fitted scenario (0.0320 vs 0.1122) and 44% in the Textbook scenario (0.1179 vs 0.2097).

8 Is σ Really a Regularizer? A Cross-Validation Check

Everything above rests on a claim that Chapter 13 makes but does not check: that σ is a regularization strength, and that the model chooses a good one. That claim is testable.

If σ really is a regularizer, then fixing it by hand at the value the model chose ought to predict new data better than fixing it anywhere else. So: refit the multilevel model twelve times with σ *pinned* at values from 0.15 to 2.5, leaving $\bar{\alpha}$ free, and score each fit by cross-validation. If the claim holds, the cross-validation curve should peak where the posterior of σ peaked in the model that estimated it.

```
sigma_grid <- exp( seq( log(0.15) , log(2.5) , length.out=12 ) )

for ( i in seq_along(sigma_grid) ) {
  dat_s <- c( dat , list( sigma_fixed=sigma_grid[i] ) )
  fits[[i]] <- ulam(
    alist(
      P ~ dbinom( N , p ),
      logit(p) <- a[listing],
      a[listing] ~ dnorm( a_bar , sigma_fixed ),
      a_bar ~ dnorm( 0 , 1.5 )
    ), data=dat_s , chains=4 , log_lik=TRUE )
}
```

8.1 What PSIS is, and why it breaks here

The idea. Leave-one-out cross-validation asks: how well would the model have predicted listing i if it had never seen listing i ? Answering that honestly means refitting the model two hundred times, once per listing, which is expensive. PSIS is the shortcut. Fit *once* on all the data, then reuse those posterior draws to imitate a fit that excluded listing i , by giving each draw a weight

$$w_i^{(s)} = \frac{1}{p(P_i | \theta^{(s)})}.$$

The logic: a draw that explained listing i very well is a draw that listing i pulled the fit toward, so play it down; a draw that explained it poorly is one the fit would have been free to visit without listing i , so play it up. Reweight, average, and you have an estimate of the held-out score for free.

Where the “PS” comes in. Those weights are a ratio of probabilities, and ratios can explode: a handful of draws can end up carrying almost all the weight, and an average dominated by three draws is not an average. So PSIS (Pareto Smoothed Importance Sampling) fits a Pareto distribution to the largest weights and replaces them with the smoothed fit. That Pareto has a shape parameter k , which measures how heavy the tail of the weights is, and it doubles as the warning light: $k > 0.7$ means the estimate for that point rests on too few draws and should not be believed.

Why it breaks here. Reweighting can only work if the with-the-point posterior and the without-the-point posterior actually overlap — you cannot reweight your way somewhere your draws never went. Here they barely overlap at all. Every listing contributes exactly one observation, so α_i is held in place by listing i and by nothing else. Delete listing i and its intercept

has no data left: its posterior widens from a tight bump to the entire population distribution $\text{Normal}(\bar{\alpha}, \sigma)$. The draws we have are clustered in the tight bump, so no reweighting of them can reproduce the wide one, and k duly goes red almost everywhere.

The exact answer instead. That same fact — α_i is unconstrained once listing i is gone — is what makes the shortcut unnecessary. If nothing is known about α_i , then predicting listing i without it is just predicting a *brand new* listing drawn from the population:

$$p(P_i | y_{-i}, \sigma) = \int \text{Binomial}(P_i | N_i, \text{logit}^{-1}(a)) \text{Normal}(a | \bar{\alpha}, \sigma) da,$$

averaged over the posterior of $\bar{\alpha}$. No importance sampling: draw $\bar{\alpha}$ from the posterior, draw a around it, average the binomial density. Its only approximation is reusing the full-data posterior of $\bar{\alpha}$ instead of refitting without listing i — and dropping one listing in two hundred moves $\bar{\alpha}$ by a fraction of its own posterior standard deviation. This is the “direct” column below.

8.2 Results: the curve and the posterior agree

σ	elpd PSIS	bad k	elpd WAIC	elpd direct
0.150	−268.0	2	−267.6	−265.7
0.194	−265.2	4	−264.5	−263.1
0.250	−260.6	3	−259.3	−260.0
0.323	−257.1	7	−254.2	−256.8
0.417	−252.3	6	−248.7	−253.9
0.539	−249.4	16	−244.1	−252.2
0.696	−247.0	27	−240.0	−252.3
0.899	−247.4	36	−238.1	−255.3
1.161	−248.2	45	−237.2	−261.8
1.499	−252.4	58	−237.7	−272.2
1.936	−257.7	72	−239.7	−286.6
2.500	−264.4	97	−241.9	−304.4

Table 5: Each row is a separate model with σ pinned. Higher elpd is better; the best value in each column is bold. “bad k ” counts the listings, out of 200, whose Pareto k exceeds 0.7 — the diagnostic that says PSIS should not be believed.

The direct cross-validation curve and the posterior land in the same place. The curve is highest at $\sigma = 0.539$, with $\sigma = 0.696$ all but tied (−252.2 against −252.3), and the posterior mode of 0.623 falls between those two adjacent grid points, comfortably inside the 89% interval [0.456, 0.786]. A purely predictive criterion, which knows nothing about priors, and a purely Bayesian one, which never simulated a held-out data point, choose the same amount of pooling: **the multilevel model found by regularization what cross-validation found by brute force.** The claim is agreement, not identity — the grid is coarse, the peak is flat, and the Exponential(1) prior pulls the mode slightly down. What the curve does say sharply is that both ends are bad: pinning σ at 0.15 costs 13 elpd, and at 2.5 it costs 52.

Why PSIS and WAIC miss the mode. Both peak at *larger* σ than the direct computation — one grid step for PSIS, three for WAIC — so both recommend *less* pooling than actually predicts well. The cause is the overlap problem from above. Neither ever really removes listing i : both score it using a posterior in which α_i had already been fitted to that very listing. The “held-out” score is therefore partly in-sample, and it is *most* in-sample when σ is large, because

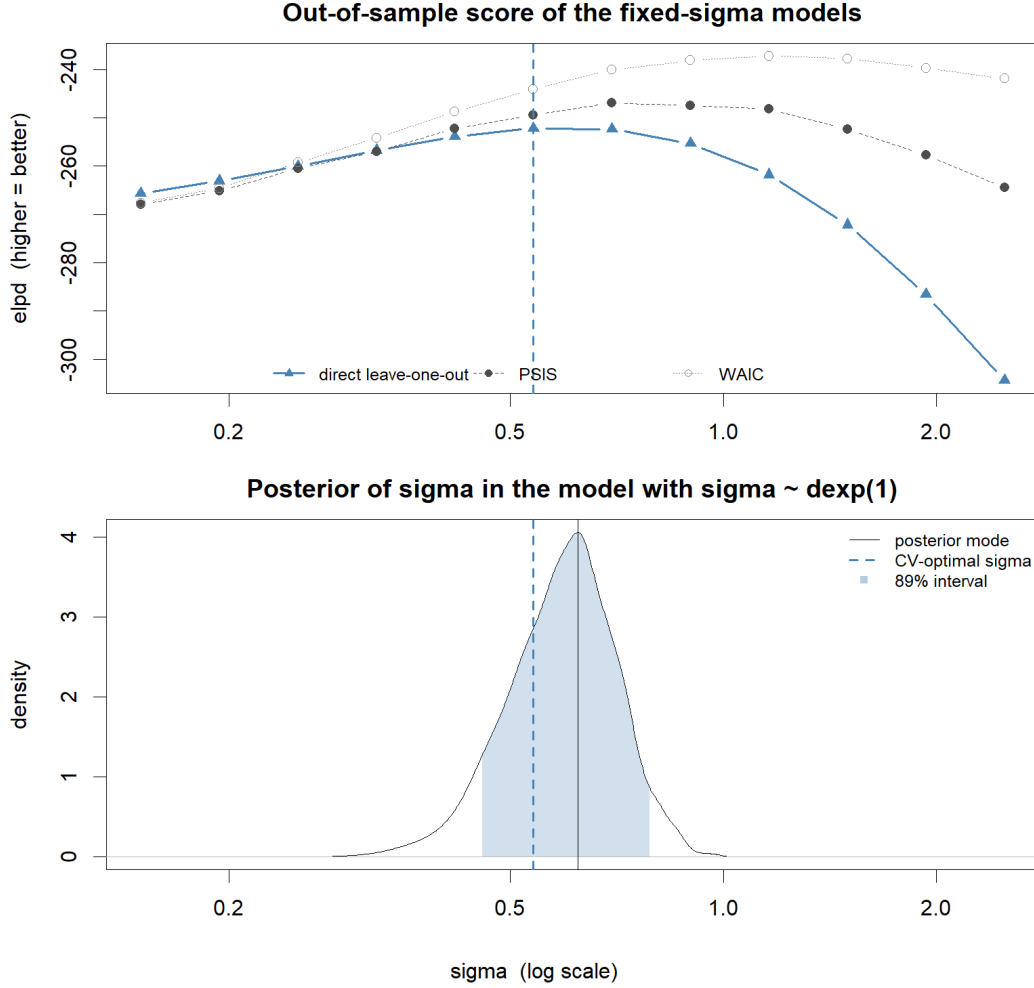


Figure 6: Top: out-of-sample score of each fixed- σ model, by three criteria. Bottom: the posterior of σ from the model that estimated it with a Exponential(1) prior, with its 89% interval shaded. The dashed line is the cross-validation optimum, carried down from the top panel.

a large σ is exactly what lets each α_i run off and match its own single observation. Larger σ thus buys a flattery the direct computation refuses to pay, and the peak slides to the right.

The bad- k column tracks the damage: it climbs from 2 at $\sigma = 0.15$ to 97 at $\sigma = 2.5$, and already stands at 27 at the σ PSIS nominates — the warning light is brightest exactly where the answer is being read off. An information criterion is an approximation, and approximations have assumptions; one observation per cluster violates this one.

9 Conclusions

Starting from the observation that a rating computed from two reviews is not the same kind of number as a rating computed from four hundred, we wrote down a generative story in which reviews are trials and ratings are proportions, and that story turned out to be the Reed frog model. Fitting it required attention to the geometry of the posterior rather than to the model itself: the reparameterization $\alpha_j = \bar{\alpha} + z_j\sigma$ changes nothing about what is being estimated and everything about how well it can be estimated.

The comparison delivered the chapter’s central result, twice over. By WAIC, the multilevel model has two more parameters than the no-pooling model and uses a third as many of them,

because it learned its own regularization instead of being handed one; the shrinkage plot showed where that regularization goes, almost entirely onto the listings with the least evidence and almost not at all onto the listings with the most. Against a simulated ground truth, where the right answer is known rather than estimated, the same ordering held: partial pooling had the lowest error in both the tight- σ and the wide- σ regime, while each of the two extremes was badly wrong in one of them. That the two criteria agree matters, because WAIC is scored on the same data the models were fitted to and the simulation is not.

Finally, pinning σ by hand and scoring the result by cross-validation confirmed that the σ the model estimates is the σ that predicts best. Along the way it also showed that the standard tools for that scoring, PSIS and WAIC, cannot be trusted when every cluster holds a single observation — which is worth knowing, because it is a common shape for data to have.

A Hamiltonian Monte Carlo

Section 4 leaned on the geometry of the posterior — trajectories, step sizes, divergences, energy — without saying where any of it comes from. That material is written up separately, in **the appendix about HMC of posterior distribution approximation**, which builds Hamiltonian Monte Carlo from the physical analogy up and explains why the funnel is hard for it and why the non-centered parameterization fixes that. It is a standalone document, nothing in this report depends on reading it first.

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